MODEL DEVELOPMENT DOCUMENT (MDD)

House Price Prediction using Machine Learning

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1. Project Overview

House price prediction is one of the most important applications of Machine Learning in the real estate industry. The price of a property depends on several factors such as location, built-up area, number of bedrooms, construction status, resale status, and geographical coordinates. Predicting house prices manually is often difficult because these factors have complex relationships that are not easy to model using traditional methods.

The objective of this project is to develop a machine learning-based regression model that accurately predicts house prices using historical housing data. The project follows a complete machine learning workflow, including data preprocessing, exploratory data analysis (EDA), feature engineering, model training, model evaluation, and hyperparameter tuning.

The dataset initially contained 29,451 records, which were reduced to 29,050 records after removing duplicate entries. Several preprocessing techniques, including One-Hot Encoding, Target Encoding, IQR-based outlier treatment, Log Transformation, and StandardScaler, were applied to improve data quality before training the models.

A total of 13 regression algorithms were implemented and evaluated using R^2 Score, Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE). Based on performance analysis, the Tuned XGBoost Regressor was selected as the final model because it achieved the highest prediction accuracy with a Test R^2 Score of

approximately 0.9071.

The trained model was saved using the Joblib library for future deployment in web-based or desktop applications, making the system suitable for practical real estate price estimation.

2. Problem Statement

House price prediction is a complex task because property prices depend on multiple factors such as location, area, number of bedrooms, construction status, and resale condition. Manual price estimation is often inaccurate and time-consuming. This project develops a machine learning-based regression model to predict house prices accurately using historical housing data. Multiple regression algorithms are trained and compared to identify the best-performing model.

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3. Objectives

The main objectives of this project are:

- To develop a machine learning model for house price prediction.
- To preprocess the dataset by removing duplicates and handling categorical features.
- To perform exploratory data analysis (EDA).
- To compare multiple regression algorithms.
- To optimize the best model using hyperparameter tuning.
- To evaluate the models using R^2 Score, MAE, and RMSE.
- To save the final trained model for future deployment.

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4. Scope of the Project

The scope of this project includes developing a house price prediction system using machine learning techniques. The project covers data preprocessing, feature engineering, model training, performance evaluation, and hyperparameter tuning. The final Tuned XGBoost model is saved for future deployment in web or desktop applications. Future improvements may include adding more property-related features and deploying the model as a real-time prediction system.

5. Dataset Description

5.1 Dataset Overview

The dataset used in this project contains information about residential properties and their corresponding prices. It includes both numerical and categorical features that influence house prices. The target variable is **TARGET (PRICE_IN_LACS)**, which represents the selling price of a property.

Initially, the dataset contained **29,451 records**. After removing **401 duplicate records**, the final dataset consisted of **29,050 unique records**, which were used for model development.

Dataset Summary

Attribute	Description
Total Records (Original)	29,451
Duplicate Records Removed	401
Final Records	29,050
Total Features	12
Target Variable	TARGET (PRICE_IN_LACS)
Problem Type	Supervised Machine Learning (Regression)

5.2 Feature Description

Feature Name	Description
POSTED_BY	Property listed by Owner, Dealer, or Builder
UNDER_CONSTRUCTION	Construction status of the property
RERA	RERA approval status
BHK_NO.	Number of bedrooms
BHK_OR_RK	Property type (BHK/RK)
SQUARE_FT	Area of the property in square feet
READY_TO_MOVE	Ready-to-move status
RESALE	Resale property status
ADDRESS	Property location
LONGITUDE	Longitude coordinate
LATITUDE	Latitude coordinate
TARGET (PRICE_IN_LACS)	House price (Target Variable)

The dataset contains both numerical and categorical attributes, making it suitable for regression-based machine learning models after appropriate preprocessing.

6. Data Preprocessing

Data preprocessing was performed to improve the quality of the dataset before training the machine learning models. Initially, the dataset was examined using Pandas functions such as info(), describe(), and isnull() to understand its structure and verify data quality. No missing values were found in the dataset.

A total of **401 duplicate records** were identified and removed, reducing the dataset size from **29,451** to **29,050** records. This ensured that each property was represented only once during model training.

Categorical variables **POSTED_BY** and **BHK_OR_RK** were converted into numerical format using **One-Hot Encoding**, while the **ADDRESS** feature was transformed using **Target Encoding** because it contained a large number of unique values. This reduced dimensionality while preserving important location-based information.

To reduce the effect of extreme values, the **Interquartile Range (IQR)** method was applied to **SQUARE_FT** and **BHK_NO.** Outliers were clipped instead of removed to retain maximum information from the dataset.

Since **SQUARE_FT**, **BHK_NO.**, and **TARGET (PRICE_IN_LACS)** showed positive skewness, **Log Transformation (log1p)** was applied to normalize their distributions and improve model performance.

The dataset was then divided into **80% training data (23,240 records)** and **20% testing data (5,810 records)** using train_test_split() with a random state of **42**. Finally, **StandardScaler** was applied to **SQUARE_FT**, **BHK_NO.**, **LONGITUDE**, and **LATITUDE** to standardize numerical features and ensure consistent scaling across all machine learning algorithms.

Preprocessing Summary

Step	Technique Used
Missing Value Check	No missing values found
Duplicate Removal	401 records removed
One-Hot Encoding	POSTED_BY, BHK_OR_RK
Target Encoding	ADDRESS
Outlier Treatment	IQR Clipping

Step	Technique Used
Log Transformation	SQUARE_FT, BHK_NO., TARGET
Train-Test Split	80:20
Feature Scaling	StandardScaler

7. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to understand the characteristics of the dataset and identify important patterns before model training. Various statistical summaries and visualizations were used to analyze the relationships between features and the target variable.

A **correlation heatmap** was generated to measure the relationship among numerical variables. The analysis showed that **SQUARE_FT** had the strongest positive correlation with **TARGET (PRICE_IN_LACS)**, indicating that larger properties generally have higher prices. **BHK_NO.** also showed a positive relationship with the target variable, while features such as **RERA** and **UNDER_CONSTRUCTION** had a comparatively lower influence.

Histograms revealed that **SQUARE_FT** and **TARGET (PRICE_IN_LACS)** were positively skewed. Therefore, logarithmic transformation was applied to normalize their distributions before model training.

Box plots were used to detect outliers in numerical features. Significant outliers were observed in **SQUARE_FT** and **BHK_NO.**, which were treated using the **Interquartile Range (IQR)** clipping method to reduce their impact on model performance.

Overall, the EDA confirmed that the dataset was suitable for machine learning after preprocessing and highlighted the importance of property size, location, and the number of bedrooms in predicting house prices.

Key Observations

Observation	Result
Missing Values	None
Duplicate Records	401 removed
Most Influential Feature	SQUARE_FT
Outliers Found	SQUARE_FT, BHK_NO.
Distribution	Positively Skewed

Observation	Result
Outlier Treatment	IQR Clipping
Transformation Applied	Log Transformation

8. Feature Engineering and Model Development

8.1 Feature Engineering

Feature engineering was performed to improve the predictive capability of the machine learning models. The categorical variables **POSTED_BY** and **BHK_OR_RK** were converted into numerical format using **One-Hot Encoding**, while the **ADDRESS** feature was encoded using **Target Encoding** to handle its high cardinality without significantly increasing the number of features.

To reduce skewness and stabilize variance, **Log Transformation (log1p)** was applied to **SQUARE_FT**, **BHK_NO.**, and **TARGET (PRICE_IN_LACS)**. Numerical features such as **SQUARE_FT**, **BHK_NO.**, **LONGITUDE**, and **LATITUDE** were standardized using **StandardScaler**, ensuring that all features were on a comparable scale before model training.

The final dataset contained both transformed numerical features and encoded categorical features, making it suitable for regression algorithms.

8.2 Model Development

After preprocessing and feature engineering, the dataset was divided into **80% training data** and **20% testing data**. Multiple regression algorithms were trained and evaluated to identify the best-performing model for house price prediction.

The following machine learning algorithms were implemented:

- Linear Regression
- Ridge Regression
- Lasso Regression
- ElasticNet Regression
- Decision Tree Regressor
- Random Forest Regressor
- Gradient Boosting Regressor
- AdaBoost Regressor

- K-Nearest Neighbors (KNN)
- Support Vector Regressor (SVR)
- XGBoost Regressor
- LightGBM Regressor
- CatBoost Regressor

Each model was evaluated using **R^2 Score**, **Mean Absolute Error (MAE)**, and **Root Mean Squared Error (RMSE)**. Based on the evaluation results, ensemble boosting models such as **XGBoost**, **CatBoost**, and **Random Forest** achieved better prediction accuracy than traditional regression models due to their ability to capture complex non-linear relationships within the housing data.

Model Development Summary

Parameter	Value
Total Models Trained	13
Problem Type	Regression
Train-Test Split	80:20
Evaluation Metrics	R^2 Score, MAE, RMSE
Best Performing Model	Tuned XGBoost
Model Saving	Joblib (house_price_model.pkl)

9. Model Evaluation and Performance Analysis

After training all regression models, their performance was evaluated using three standard metrics: **R^2 Score**, **Mean Absolute Error (MAE)**, and **Root Mean Squared Error (RMSE)**. These metrics were used to compare prediction accuracy and identify the best-performing model.

- **R^2 Score** measures how well the model explains the variation in house prices. A value closer to **1** indicates better performance.
- **MAE** measures the average absolute prediction error. Lower values indicate higher accuracy.
- **RMSE** measures the overall prediction error while giving higher importance to larger errors. Lower RMSE indicates better model performance.

Performance Comparison of Regression Models

Model	Test R^2	Test MAE	Test RMSE
Linear Regression	0.6645	0.3496	0.5136
Ridge Regression	0.6645	0.3496	0.5136
Lasso Regression	0.6639	0.3493	0.5141
ElasticNet	0.6642	0.3493	0.5139
Decision Tree	0.8203	0.2333	0.3759
Random Forest	0.8928	0.1802	0.2904
Gradient Boosting	0.8517	0.2350	0.3415
AdaBoost	0.6187	0.4016	0.5476
K-Nearest Neighbors	0.8042	0.2586	0.3924
Support Vector Regressor	0.7297	0.3011	0.4611
XGBoost	**0.8984**	**0.1848**	**0.2826**
LightGBM	0.8873	0.2000	0.2977
CatBoost	0.8947	0.1903	0.2878

Performance Analysis

The results indicate that ensemble learning algorithms significantly outperformed traditional regression models. Linear Regression, Ridge Regression, Lasso Regression, and ElasticNet achieved Test R^2 values of approximately **0.66**, indicating that linear models were unable to capture the complex relationships within the housing dataset.

Decision Tree Regression achieved a higher Test R^2 Score (**0.8203**) but showed signs of overfitting due to its tendency to memorize training data. Random Forest improved prediction accuracy by combining multiple decision trees and achieved a Test R^2 Score of **0.8928** with a lower RMSE (**0.2904**).

Among all baseline models, **XGBoost** produced the best performance with a **Test R^2 Score of 0.8984**, **MAE of 0.1848**, and **RMSE of 0.2826**, demonstrating its ability to learn complex non-linear relationships while maintaining strong generalization. CatBoost (**0.8947**) and LightGBM (**0.8873**) also achieved competitive performance, confirming that boosting-based algorithms are well suited for house price prediction.

Based on these results, **XGBoost** was selected for further optimization through hyperparameter tuning because it achieved the highest prediction accuracy among all evaluated baseline models.

10. Hyperparameter Tuning and Final Model Selection

10.1 Hyperparameter Tuning

After comparing the performance of all regression models, **XGBoost Regressor** was selected for further optimization because it achieved the highest baseline performance. Although the default XGBoost model produced excellent results, its prediction accuracy was further improved through **RandomizedSearchCV**.

RandomizedSearchCV was preferred over GridSearchCV because it explores a wide range of parameter combinations with lower computational cost while providing near-optimal results. A **5-fold Cross Validation** strategy was used to ensure that the selected parameters generalized well to unseen data.

The tuning process optimized important hyperparameters such as:

- Number of Estimators (n_estimators)
- Learning Rate (learning_rate)
- Maximum Tree Depth (max_depth)
- Minimum Child Weight (min_child_weight)
- Gamma (gamma)
- Subsample Ratio (subsample)
- Column Sampling Ratio (colsample_bytree)

These parameters directly control model complexity, learning speed, and overfitting.

10.2 Performance After Hyperparameter Tuning

After tuning, the XGBoost model showed an improvement in prediction accuracy and a reduction in prediction error.

Final Performance of Tuned XGBoost

Performance Metric	Value
Training R^2 Score	**0.9666**
Testing R^2 Score	**0.9071**

Performance Metric	Value
Training MAE	**0.116**
Testing MAE	**0.173**
Training RMSE	**0.161**
Testing RMSE	**0.270**
Cross Validation R^2	**0.9024**

The Test R^2 Score improved from **0.8984** (baseline XGBoost) to **0.9071** after hyperparameter tuning, indicating better predictive performance. Similarly, the Test RMSE decreased from **0.2826** to **0.270**, confirming that the optimized model produced more accurate predictions with lower overall error.

10.3 Final Model Selection

Based on the comparative analysis and hyperparameter tuning results, the **Tuned XGBoost Regressor** was selected as the final model for house price prediction.

The selection was based on the following observations:

- It achieved the **highest Test R^2 Score (0.9071)** among all evaluated models.
- It recorded the **lowest prediction errors (MAE = 0.173, RMSE = 0.270)**.
- The difference between **Training R^2 (0.9666)** and **Testing R^2 (0.9071)** was relatively small, indicating good generalization and limited overfitting.
- Cross-validation also produced a high R^2 Score (**0.9024**), demonstrating that the model performs consistently on unseen data.

The trained model and preprocessing objects were saved using the **Joblib** library for future deployment.

Saved Files

File Name	Purpose
house_price_model.pkl	Trained Tuned XGBoost model
scaler.pkl	StandardScaler object used during preprocessing
target_encoder.pkl	Target Encoder used for the ADDRESS feature

These files can be loaded directly during deployment to ensure that new input data undergoes the same preprocessing steps before prediction. This makes the prediction pipeline consistent, reproducible, and suitable for integration into web applications or other real-world systems.

11. Conclusion and Future Scope

11.1 Conclusion

This project successfully developed a **Machine Learning-based House Price Prediction System** using a structured housing dataset. A complete machine learning pipeline was implemented, beginning with data preprocessing and ending with the selection of the best-performing regression model.

Initially, the dataset was cleaned by removing duplicate records, followed by categorical feature encoding using **One-Hot Encoding** and **Target Encoding**. Outliers were handled using the **Interquartile Range (IQR)** method, while **Log Transformation** was applied to reduce skewness in important numerical features. Finally, feature scaling was performed using **StandardScaler** to improve the learning efficiency of machine learning algorithms.

A total of **13 regression algorithms** were trained and evaluated using **R^2 Score, Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE)**. Among all models, the **Tuned XGBoost Regressor** achieved the highest prediction accuracy with a **Test R^2 Score of 0.9071**, indicating that the model successfully explained approximately **90.71%** of the variation in house prices.

The trained model was saved using the **Joblib** library (house_price_model.pkl) along with the preprocessing objects (scaler.pkl and target_encoder.pkl). These files ensure that the prediction pipeline remains consistent during deployment.

Overall, the developed model demonstrates high prediction accuracy, good generalization capability, and can effectively assist buyers, sellers, and real estate professionals in estimating property prices.

Machine Learning Workflow

Dataset Collection → Data Cleaning → Duplicate Removal → Encoding → Outlier Treatment → Log Transformation → Feature Scaling → Train-Test Split → Model Training → Model Evaluation → Hyperparameter Tuning → Final Model Selection → Model Saving → DeploymentThe workflow ensures that every stage of the machine learning lifecycle is systematically executed, resulting in a reliable and reproducible house price prediction model.